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Functional outcomes in schizophrenia: a comparison of olanzapine and haloperidol in a European sample

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The primary aim of this study was to compare functional outcomes between patients with schizophrenia treated with olanzapine or haloperidol in Europe. The sample consisted of European patients from a large, international, double-blind, randomized clinical trial. Patients were randomized to receive either olanzapine ($n = 520$) or haloperidol ($n = 258$) for a 6-week acute phase followed by a 46-week maintenance phase for responders. Olanzapine-treated patients experienced superior improvements compared to haloperidol-treated patients on all efficacy measures assessed in both phases. A greater percentage of olanzapine-treated patients had ≥ 20% improvement in the Quality of Life Scale total score during both the acute (50.0% versus 31.0%, $P = 0.071$) and maintenance (69.5% versus 41.7%, $P = 0.006$) phases compared to haloperidol-treated patients. For patients who entered the maintenance phase as outpatients, olanzapine-treated patients were significantly less likely to require subsequent hospitalization compared to haloperidol-treated patients ($P = 0.001$). A significantly greater percentage of the olanzapine group compared to the haloperidol group worked part-time or full-time (15.1% versus 5.3%, $P = 0.018$), participated in useful work ≥ 75% of the time (21.0% versus 10.5%, $P = 0.038$), and socialized more than once a month (53.8% versus 37.9%, $P = 0.004$) during the maintenance phase. The findings from this study suggest that olanzapine's clinical profile leads to reduced hospitalization and improvements in work and social functioning superior to that achieved with haloperidol treatment. Int Clin Psychopharmacol 15:245–255 © 2000 Lippincott Williams & Wilkins

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INTRODUCTION

Schizophrenia afflicts approximately 1% of the population and is the most incapacitating psychiatric disorder (Keith *et al.*, 1991). Despite the availability of treatment options in community settings, patients with schizophrenia often require hospitalization. Furthermore, at least 50% of patients are readmitted within 2 years after recovery from a first episode of schizophrenia (Shepherd *et al.*, 1989; Scottish Schizophrenia Research Group, 1993; Weiden and Olfson, 1995; Knapp and Kavanagh, 1997). Patients may have anxiety, depression, lack of volition and motivation, little insight, and cognitive impairment. They are frequently withdrawn socially, and have difficulty forming close relationships with others (Crismon and Dorson, 1996). Therefore, independent living and gainful employment are difficult for people with schizophrenia.

Since the introduction of chlorpromazine in the 1950s, conventional antipsychotic medications have become the most common treatment in both the acute and maintenance phases of schizophrenia. While conventional antipsychotics reduce positive symptoms (Davis *et al.*, 1989; Baldessarini *et al.*, 1990; Dixon *et al.*, 1995; Kane, 1996), they are only modestly effective in improving negative symptoms (Dixon *et al.*, 1995). Furthermore, many patients treated with conventional antipsychotics develop extrapyramidal side-effects (EPS) (Tarsay, 1983), which in turn may exacerbate negative symptoms (Davis *et al.*, 1989; Baldessarini *et al.*, 1990).

The impact of antipsychotic treatment on negative

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symptoms is crucial in that negative symptoms are important predictors of social and vocational outcome (Meltzer *et al.*, 1990; Breier *et al.*, 1991; Hwu *et al.*, 1995; Lehman, 1996; Marder, 1996; Davidson and McGlashan, 1997). While both positive and negative symptom levels are predictive of outcome, negative symptoms are more chronic and are more likely to be associated with neurocognitive impairment and work and social outcomes (Davidson and McGlashan, 1997). Negative symptoms and cognitive impairment decrease life satisfaction and impair functional status more than positive symptoms (Melter *et al.*, 1990). Therefore, antipsychotic agents that are effective in reducing negative symptoms can have a greater impact on quality of life and social and vocational function.

While atypical antipsychotics have been shown to be more effective in controlling negative symptoms (Kane *et al.*, 1988; Fitton and Heel, 1990; Meltzer *et al.*, 1990; Breier *et al.*, 1994; Carman *et al.*, 1995; Borrison, 1997), information on their impact on social and work outcomes is limited. Clozapine used in patients who continued treatment for at least 2 years (compared to a dropout group) led to improvements in work functioning, capacity for independent living, and re-hospitalization rates (Meltzer *et al.*, 1990). Research has found that compared to haloperidol-treated patients, clozapine-treated patients had fewer mean days of hospitalization for psychiatric reasons, and that among patients who completed their assigned treatment, clozapine therapy resulted in significant improvement in quality of life (Rosenheck *et al.*, 1997). Additional research demonstrated that patients randomized to risperidone had statistically superior improvement in mental health functioning on the SF-36 Health Survey compared to patients randomized to conventional treatment (Meredith *et al.*, 1998). However, results from the Heinrichs and Carpenter Quality of Life Scale (QLS) (Heinrichs *et al.*, 1984) showed no difference between the two treatment groups.

Olanzapine was found to be more effective than haloperidol in reducing psychopathology and negative symptoms, and had greater improvements in quality of life (Beasley *et al.*, 1996; Tollefson *et al.*, 1997; Hamilton *et al.*, 1998; Revicki *et al.*, 1999). To provide increased relevance to European audiences and to reduce variation associated with cultural differences as well as differences in health care and social service systems across regions (e.g. Europe versus North America), outcomes were assessed in the European subset of patients with schizophrenia enrolled in an international clinical trial.

METHODS

Design and patient sample

The parent study for this analysis was a double-blind, randomized clinical trial including 174 clinical centres in 17 countries. A total of 1996 inpatients or outpatients with a documented DSM-III-R diagnosis of schizophrenia (American Psychiatric Association, 1987), schizophreniform disorder, or schizoaffective disorder were enrolled (Tollefson *et al.*, 1997). Patients were at least 18 years old and either had a Brief Psychiatric Rating Scale (BPRS) (Woerner *et al.*, 1988) total score (items scored on a 0–6 scale) of ≥ 18 and/or were no longer tolerating current neuroleptic (excluding haloperidol) therapy. Reasons for exclusion included a documented treat-

Table 1. Sample size by country

Country	Acute phase		Maintenance phase	
	Olanzapine	Haloperidol	Olanzapine	Haloperidol
Austria	16	7	14	4
Belgium	44	22	34	4
Denmark	11	5	9	3
Finland	22	11	10	5
France	204	104	141	48
Germany	45	21	26	7
Italy	26	14	15	5
Luxembourg	3	2	2	2
Netherlands	3	2	2	1
Portugal	5	3	5	2
Spain	36	17	26	7
Sweden	10	6	2	3
UK	95	44	33	13
Total	520	258	319	104

ment-resistance to neuroleptic agents, DSM-III-R organic mental disorder or substance-use disorder, and/or a serious, unstable medical illness.

For this analysis, the sample was restricted to patients recruited from 96 clinical centres located in the 13 European countries of the parent study (n = 778, 39% of total study sample). The distribution of patients by country is shown in Table 1.

Treatment regimen

After an antipsychotic washout period, patients were randomized in a 2 : 1 ratio to receive olanzapine or haloperidol both at 5 and 20 mg/day for the 6-week acute phase. The initial 5 mg/day dose was increased weekly for patients whose Clinical Global Impression (CGI) Severity (Guy, 1976) score was > 1. Dose decreases could occur at any time. Only benzodiazepine for sedation and benzotropine or biperiden to control EPS were allowed as psychotropic concomitant medications.

After completing the acute phase, patients showing a predefined level of response (maximum CGI Severity score of 3 or a decrease in score of ≥ 3, a maximum CGI Severity of Adverse Events score of ≥ 3, and clinician judgment that continued treatment was warranted) were eligible for continued double-blind therapy in the 46-week maintenance phase.

Assessments

The primary clinical outcome assessed in this trial was the BPRS extracted from the Positive and Negative Syndrome Scale (PANSS) (Guy, 1976; Kay *et al.*, 1986). The PANSS was administered weekly during the acute phase and every 8 weeks during the maintenance phase. A clinically important response was defined prospectively in the protocol as a 40% improvement from baseline.

Hospitalization data (admission and discharge date, and the primary ICD-9-CM diagnosis associated with the hospitalization) (International Classification of Diseases, 1993) and social and vocational outcome data were collected at the end of the acute phase, at 8-week intervals throughout the maintenance phase, and at the final visit. Hospitalization data were verified by hospital records. Social and vocational outcomes were assessed with the QLS as well as with items adapted from the Level of Function Scale (Strauss and Carpenter, 1974).

The QLS is designed to evaluate the current functioning of people with schizophrenia apart from the presence or absence of florid psychotic symptomatology or need for hospitalization (Heinrichs *et al.*, 1984). The instrument assesses the richness of the patient's personal experience, the quality of their interpersonal relations, and their productivity in occupational roles. The QLS contains 21 questions rated by clinicians on 7-point scales following a semi-structured interview with the patient. A QLS total and four subscale scores are calculated, with higher scores indicating less impairment. The four subscale scores are: Intrapsychic Foundations, Interpersonal Relations, Instrumental Role Category, and Common Objects and Activities. Intrapsychic Foundations evaluates the patient's sense of purpose, motivation, curiosity, empathy, ability to experience pleasure, and emotional interaction. Interpersonal Relations measures different aspects of social relations and interpersonal experience, including capacity for intimacy, active versus passive participation in events, and avoidance and withdrawal tendencies. Instrumental Role Category focuses on various social roles, such as student, housekeeper, or worker, as well as level of accomplishment and satisfaction derived from these roles. Within the Common Objects and Activities subscale, patients are asked if they are wearing or carrying common items (e.g. wallet, watch, credit card) and engaging in a range of regular activities (e.g. read a newspaper, pay a bill). The assumption underlying this scale is that the absence of common objects and/or failure to engage in common activities reflects impairment in day-to-day life participation. The QLS has good inter-rater and intrarater reliability and evidence supporting construct and convergent validity (Heinrichs *et al.*, 1984; Lehman *et al.*, 1993). The QLS has been used in several clinical studies of antipsychotic medications (Meltzer *et al.*, 1990; Breier *et al.*, 1994; Beasley *et al.*, 1996; Rosenheck *et al.*, 1997; Revicki *et al.*, 1999).

A clinically important response on the QLS was defined as improvement of at least 20% from baseline (Rosenheck *et al.*, 1997). The QLS was administered to mainly outpatients in English-, German-, French-, Italian-, and Spanish-speaking countries only. Thus, a subset of patients from Austria, Belgium, France, Germany, Italy, Spain and UK were included in QLS analyses.

Work and social outcome items adapted from the Level of Function Scale included work status, amount of useful work, social relations, and living arrangements and were based on the patient's status for the majority of the time during the assessment period. Patients were classified from unemployed to full-time work for pay based on clinician ratings for the work status outcome. Amount of time involved in functio-

nal activities/work (includes working for pay, student, housekeeping, and volunteer work) was also assessed. The social relations question assessed patients' social activities ranging from never visits friends and relatives to visits friends and relatives more than once a week. Patients were classified as to whether or not they had significant social interaction more than once per month. The living arrangements item assessed where patients were residing for the majority of time during the assessment period. Options ranged from independent living (living in own apartment or house) to homeless. The data were gathered from one or more of the following sources: all available medical records, psychiatric history, and direct questions to patients and/or their family members.

Statistical methods

All analyses were carried out on the intent-to-treat population, meaning all patients were included in the groups to which they were randomly assigned, even if a patient did not strictly adhere to the protocol. All endpoint analyses used a last observation carried forward (LOCF) algorithm; that is, the last available visit served as endpoint. Main effects were tested at $\alpha = 0.05$ and interactions and heterogeneity among countries were tested at $\alpha = 0.10$. SAS was used to perform all statistical analyses (SAS Institute, Inc, 1990).

Baseline demographic characteristics, clinical and quality of life data, and completion and discontinuation rates were summarized by treatment group for patients who entered the acute phase and for patients who entered the maintenance phase. Fisher's exact test was used to evaluate treatment group differences in categorical variables. Analysis of variance (ANOVA) was used for continuous variables.

For continuous clinical, social, and vocational outcome parameters, ANOVA was used to assess treatment effects between the olanzapine and haloperidol groups. The ANOVA model contained the terms treatment, country, and the interaction. Additionally, an analysis of covariance (ANCOVA) was used to assess differences between the treatment groups in clinical effects while adjusting for significant differences between treatment groups in baseline characteristics. The ANOVA model used to evaluate treatment group differences in the QLS did not contain the interaction term because of the reduced sample size in the analysis. The Cochran-Mantel-Haenszel statistic, stratified by country, was used to assess treatment group differences in BPRS and PANSS response rates ($\geq 40\%$ improvement from baseline) and QLS response rates ($\geq 20\%$ improvement from baseline). Only patients with baseline BPRS total scores ≥ 18 were included in the BPRS and PANSS response rate analyses. Similarly, only patients with QLS total and subscale scores > 0 were included in the QLS response rate analyses. Heterogeneity among countries was evaluated using the Breslow–Day statistic. Fisher's exact test was used to analyse treatment group differences within countries when significant heterogeneity was present.

For patients who entered the maintenance phase as outpatients, length of time maintaining outpatient status was estimated for each treatment group using Kaplan–Meier survival analysis. A treatment group comparison of time maintaining outpatient status was performed using the log-rank test.

Long-term work and social function outcomes were compared between treatment groups for patients who entered the maintenance phase. Baseline for this analysis was defined as the end of the acute phase (week 6 of treatment). The Cochran–Mantel–Haenszel statistic, stratified by country, was used to assess treatment group differences.

RESULTS

Baseline characteristics and patient disposition

Acute phase analyses were based on data from 778 patients (520 randomized to olanzapine and 258 randomized to haloperidol). Of these patients, 423 continued on to the maintenance phase (319 olanzapine-treated and 104 haloperidol-treated patients).

At baseline patients' mean $\pm$ SD age was 38 ± 12, 61.1% were male, and 91.9% were Caucasian. The majority of the patients had chronic schizophrenia (88.7%), were inpatients (62.7%), and had been hospitalized at some point during the 2-year period prior to beginning the study (70.4%). The mean $\pm$ SD duration of illness was 13.4 ± 10.8 years. Compared to haloperidol-treated patients, olanzapine-treated patients had significantly lower mean baseline BPRS total (34.1 ± 10.9 versus 35.8 ± 11.4, $P = 0.023$) and PANSS negative scores (25.1 ± 6.7 versus 26.0 ± 6.9, $P = 0.033$), although numerical differences were small. There were no other statistically significant baseline differences by treatment group. Overall, the mean $\pm$ SD baseline PANSS positive score was 21.6 ± 6.2. The mean $\pm$ SD baseline QLS total score was 49.5 ± 19.0.

Patients continuing into the maintenance phase had similar baseline characteristics compared to patients in the acute phase, with the exception that

Table 2. Mean change from baseline to LOCF endpoint in efficacy scores: acute and maintenance phases

Variable	Olanzapine		Haloperidol		ANOVA *P*-value[a]	ANCOVA *P*-value[b]
	n	Mean ± SD	*n*	Mean ± SD		
Acute phase (weeks 1–6)[c]						
BPRS total	514	−12.7 ± 14.3	255	−9.8 ± 13.4	0.229	0.034
PANSS positive	514	−5.7 ± 7.3	255	−4.7 ± 6.5	0.245	0.081
PANSS negative	513	−5.4 ± 6.7	255	−4.0 ± 6.8	0.300	0.035
Maintenance phase (weeks 7–52)[c]						
BPRS total	319	−16.5 ± 14.4	104	−14.7 ± 16.3	0.760	0.154
PANSS positive	319	−7.5 ± 7.8	104	−6.5 ± 7.7	0.271	0.094
PANSS negative	319	−7.2 ± 7.3	104	−6.8 ± 8.0	0.549	0.727

[a]ANOVA model including treatment, country, and treatment group by country interaction. [b]Acute phase ANCOVA model including treatment, baseline Brief Psychiatric Rating Scale (BPRS) total, baseline Positive and Negative Syndrome Scale (PANSS) negative, country, and treatment group by country interaction. Maintenance Phase ANCOVA model including treatment, baseline BPRS total, baseline PANSS negative, duration of illness, country, and the treatment group by country interaction. [c]Baseline is week 0 (randomization).

the mean duration of illness was significantly shorter in the olanzapine group compared to the haloperidol group (12.7 ± 10.9 years versus 15.2 ± 11.1 years, $P = 0.017$).

Acute phase completion rates were significantly greater for the olanzapine group compared to the haloperidol group (69.4% versus 53.9%; $P < 0.001$). The most common reason for discontinuation was lack of efficacy (olanzapine 17.9% versus haloperidol 23.3%, $P = 0.085$). A significantly smaller percentage of olanzapine-treated compared to haloperidol-treated patients discontinued for an adverse event (3.7% versus 8.9%, $P = 0.004$) and patient decision (3.5% versus 7.8%, $P = 0.013$). Other reasons for discontinuation included loss to follow-up, criteria not met, and sponsor decision. Few patients discontinued for these reasons.

During the maintenance phase, completion rates were significantly greater in the olanzapine group compared to the haloperidol group (52.0% versus 35.6%, $P = 0.005$). Significantly smaller percentages of olanzapine-treated compared to haloperidol-treated patients discontinued the maintenance phases due to adverse events (12.9% versus 26.0%, $P = 0.003$) and patient decision (10.7% versus 19.2%, $P = 0.028$). Discontinuation rates for lack of efficacy were lower in the maintenance phase compared to the acute phase, but did not differ significantly between treatment groups (olanzapine 12.5% versus haloperidol 10.6%, $P = 0.729$). Few patients discontinued for loss to follow-up, criteria not met, and sponsor decision.

Clinical outcomes

Mean changes from baseline to LOCF endpoint in efficacy scores for the acute and maintenance phases are shown in Table 2. Olanzapine-treated patients had greater mean improvements on all of the efficacy measures assessed in the acute phase compared to haloperidol-treated patients. No significant acute

Table 3. Efficacy response[a] rates: acute and maintenance phases

	Olanzapine		Haloperidol		
	n[b]	Number improved (%)	*n*[b]	Number improved (%)	*P*-value[c]
Acute phase (weeks 1–6)					
BPRS total	495	265 (53.5)	251	88 (35.1)	< 0.001
PANSS positive	495	182 (36.8)	251	64 (25.5)	0.001
PANSS negative	494	110 (22.3)	251	29 (11.6)	< 0.001
Maintenance phase (weeks 7–52)					
BPRS total	308	200 (64.9)	102	57 (55.9)	0.067
PANSS positive	308	150 (48.7)	102	45 (44.1)	0.327
PANSS negative	308	106 (34.4)	102	27 (26.5)	0.095

[a]Response defined as ≥ 40% improvement from baseline (week 0). [b]Baseline Brief Psychiatric Rating Scale (BPRS) total score ≥ 18. [c]Cochran-Mantel-Haenszel test stratified by country. PANSS, Positive and Negative Syndrome Scale.

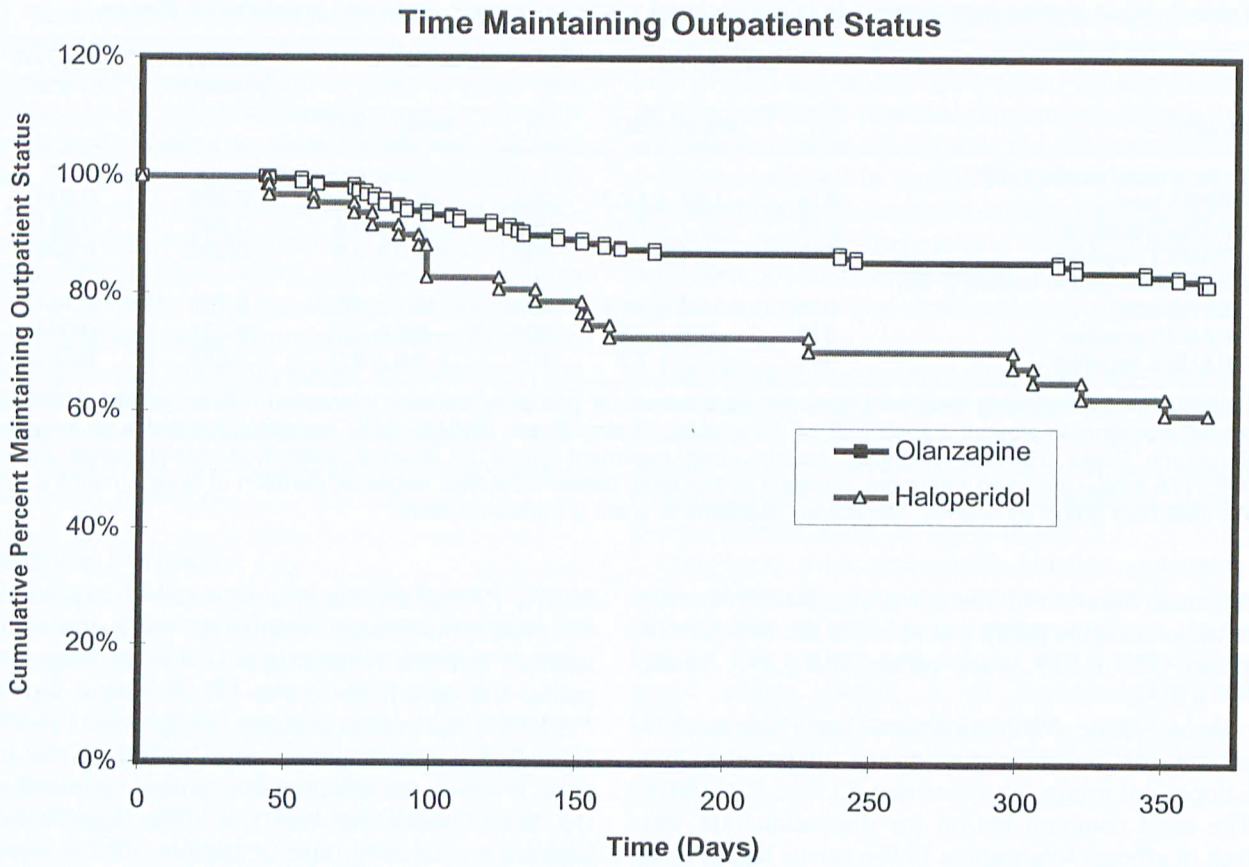


Figure 1. Time to first hospitalization: maintenance phase.

Table 4. Mean change from baseline to LOCF endpoint in QLS scores: acute and maintenance phases

	Olanzapine		Haloperidol		
	n^a	Mean ± SD	n^a	Mean ± SD	*P*-value[b]
Acute phase (weeks 1–6)					
QLS — Total	130	10.3 ± 15.8	58	5.4 ± 17.4	0.225
QLS — Intrapsychic foundations	130	3.8 ± 6.4	58	1.8 ± 6.4	0.368
QLS — Interpersonal relations	130	4.2 ± 6.5	58	1.7 ± 7.5	0.057
QLS — Instrumental role category	130	1.2 ± 4.7	58	1.7 ± 4.1	0.418
QLS — Common objects and activities	130	1.1 ± 2.0	58	0.2 ± 2.1	0.024
Maintenance phase (weeks 7–52)					
QLS — Total	105	19.1 ± 18.7	36	7.6 ± 20.6	0.007
QLS — Intrapsychic foundations	105	6.4 ± 7.4	36	1.7 ± 7.2	0.004
QLS — Interpersonal relations	105	7.0 ± 7.7	36	3.7 ± 8.9	0.059
QLS — Instrumental role category	105	4.0 ± 5.3	36	1.5 ± 5.6	0.020
QLS — Common objects and activities	105	1.7 ± 2.4	36	0.7 ± 2.4	0.128

[a]Baseline is week 0 (randomization). Scale collected mainly for outpatients. [b]ANOVA model including treatment and country.

Table 5. QLS response[a] rates: acute and maintenance phases

	Olanzapine		Haloperidol		
	n[b]	Number improved (%)	n[b]	Number improved (%)	P-value[c]
Acute phase (weeks 1–6)					
QLS — Total	130	65 (50.0)	58	18 (31.0)	0.071
QLS — Intrapsychic foundations	129	55 (42.6)	58	21 (36.2)	0.865
QLS — Interpersonal relations	130	69 (53.1)	58	19 (32.8)	0.035
QLS — Instrumental role category	115	44 (38.3)	48	15 (31.3)	0.595
QLS — Common objects and activities	127	48 (37.8)	58	17 (29.3)	0.650
Maintenance phase (weeks 7–52)					
QLS — Total	105	73 (69.5)	36	15 (41.7)	0.006
QLS — Intrapsychic foundations	104	68 (65.4)	36	12 (33.3)	0.002
QLS — Interpersonal relations	105	75 (71.4)	36	13 (36.1)	0.001
QLS — Instrumental role category	92	56 (60.9)	31	15 (48.4)	0.153
QLS — Common objects and activities	103	61 (59.2)	36	17 (47.2)	0.304

[a]Response defined as $\geq 20\%$ improvement from baseline (week 0). [b]Baseline QLS score > 0. [c]Cochran–Mantel–Haenszel test stratified by country.

phase differences in mean change from baseline to LOCF endpoint were found on any of the efficacy measures assessed when no adjustments were made for significant baseline differences between treatment groups. However, after adjusting for baseline BPRS total and PANSS negative scores in the ANCOVA, olanzapine-treated patients had significantly greater mean improvement in both BPRS total and PANSS negative scores compared to haloperidol-treated patients. In the maintenance phase, olanzapine-treated patients had greater mean improvements on all of the efficacy measures assessed, but these differences were not significant.

Efficacy response rates in the acute and maintenance phases are shown in Table 3. A significantly greater percentage of olanzapine-treated patients achieved a response of at least 40% improvement in BPRS total ($P < 0.001$), PANSS positive ($P = 0.001$), and PANSS negative ($P < 0.001$) scores compared to haloperidol-treated patients during the acute phase.

Hospitalizations

Time maintaining outpatient status during the maintenance phase is shown in Fig. 1. Survival analysis assessing time maintaining outpatient status for patients who entered the maintenance phase as outpatients revealed that olanzapine-treated patients were significantly ($P = 0.001$) more likely to maintain their outpatient status during maintenance treatment compared to haloperidol-treated patients.

Social and vocational outcomes

Mean changes from baseline to LOCF endpoint in QLS scores for the acute and maintenance phases are shown in Table 4. Olanzapine-treated patients had significantly greater mean improvement compared to haloperidol-treated patients in the Common Objects and Activities subscale ($P = 0.024$) of the QLS during the acute phase. No other significant treatment group differences were found on other QLS scores during the acute phase. During the maintenance phase, olanzapine-treated patients had greater functional improvements corresponding to significant mean increases in the QLS Total score ($P = 0.007$), the Intrapsychic Foundations ($P = 0.004$), and the Instrumental Role Category ($P = 0.020$) subscale scores compared to haloperidol-treated patients.

QLS response rates in the acute and maintenance phases are shown in Table 5. A significantly greater percentage of olanzapine-treated compared to haloperidol-treated patients had at least 20% improvement in the QLS Interpersonal Relations subscale ($P = 0.035$) during the acute phase. During the maintenance phase, a significantly greater percentage of olanzapine-treated compared to haloperidol-

Table 6. Vocational and social functioning: maintenance phase

		Baseline (week 6) Number responding			Endpoint (weeks 7–52) Number responding		
	n	Yes	%	*P*-value[a]	Yes	%	*P*-value[a]
Work full or part-time?							
Olz	305	28	9.2	0.705	46	15.1	0.018
Hal	95	7	7.4		5	5.3	
Useful work 75–100%?							
Olz	305	40	13.1	0.266	64	21.0	0.038
Hal	95	8	8.4		10	10.5	
Socialize > 1/month?							
Olz	305	151	49.5	0.302	164	53.8	0.004[b]
Hal	95	42	44.2		36	37.9	
Independent living?							
Olz	305	83	27.2	0.035	102	33.4	0.193
Hal	95	17	17.9		27	28.4	

[a]Cochran–Mantel–Haenszel test stratified by country. [b]Breslow Day test $P=0.005$.

treated patients had at least 20% improvement in the QLS total score ($P = 0.006$), Intrapsychic Foundations ($P = 0.002$), and Interpersonal Relations ($P = 0.001$) scores.

Assessments of baseline (entry into the maintenance phase) and long-term work and social function are summarized in Table 6. At endpoint, a significantly greater percentage of olanzapine-treated patients worked full or part-time compared to haloperidol-treated patients (15.1% versus 5.3%, $P = 0.018$). The percentage of patients working in the haloperidol group decreased during the maintenance phase. The percentage of olanzapine-treated patients participating in useful work 75–100% of the time was also significantly greater compared to haloperidol-treated patients (21.0% versus 10.5%, $P = 0.038$). The percentage of patients socializing more than once per month was significantly greater compared to haloperidol-treated patients (53.8% versus 37.9%, $P = 0.004$). This result varied by country (Breslow Day $P = 0.005$). Compared to haloperidol-treated patients, a significantly greater percentage of olanzapine-treated patients in Portugal socialized more than once per month (100% versus 0%; $P = 0.048$). No other significant treatment group differences were found by country.

DISCUSSION

To provide increased relevance to European audiences and to reduce variation associated with cultural differences as well as differences in health care and social service systems across regions (e.g. Europe versus North America), outcomes were assessed in the European subset of patients with schizophrenia enrolled in an international clinical trial.

Acute phase comparisons with haloperidol-treated patients revealed that olanzapine-treated patients achieved clinically important and statistically significant improvements in overall psychopathology, as well as in positive and negative symptoms. These findings were consistent with findings in the full sample (Tollefson *et al.*, 1997).

Furthermore, after 6 weeks of therapy, olanzapine-treated patients experienced greater differential improvements in day-to-day life participation as measured by the Common Objects and Activities Scale of the QLS compared to haloperidol-treated patients. The percentage of patients who had clinically important improvements in interpersonal functioning after 6 weeks of treatment was also significantly greater for olanzapine-treated patients compared to haloperidol-treated patients.

No significant differences between treatment groups in clinical endpoints were found in the maintenance phase. However, time to first hospitalization for outpatients was significantly longer for the olanzapine group compared to the haloperidol group. Despite there being no significant difference in clinical endpoints, assessment of overall, social, and vocational functioning in the maintenance phase revealed that olanzapine-treated patients experienced significant and clinically meaningful improvements in mental, social, vocational and overall functioning compared to haloperidol-treated patients as measured by the QLS and items adapted from the Level of Function Scale. The percentage of patients who socialized with friends and family more than once per month was significantly greater in the olanzapine

group compared to the haloperidol group. Furthermore, olanzapine-treated patients were more likely to work full-time or part-time and/or participate in useful work 75–100% of the time. These findings were consistent with findings from the full sample (Grainger *et al.*, 1998; Revicki *et al.*, 1999), and the US sample (Hamilton *et al.*, 1999) and suggest that olanzapine treatment may be more effective than conventional antipsychotics for reintegrating patients into the community.

Acute and maintenance phase findings in general and acute phase findings in particular support evidence from other studies that links negative symptom improvement with functional outcomes (Meltzer *et al.*, 1990; Breier *et al.*, 1991; Hwu *et al.*, 1995; Lehman, 1996; Marder, 1996; Davidson and McGlashan, 1997). In addition, these findings are consistent with the findings of Rosenheck *et al.* (1997) and Meredith *et al.* (1998) who found patients on clozapine and risperidone, respectively, to achieve greater levels of functioning than patients on conventional antipsychotics. These social and vocational outcomes findings are also consistent with analyses using the full sample where olanzapine was more effective than haloperidol in improving health-related quality of life (Revicki *et al.*, 1997).

When interpreting social and vocational outcome results, consideration should be given to the patient population assessed. Haloperidol-treated patients in the maintenance phase were potentially a more 'select' group of patients compared to the olanzapine-treated patients since only 40% of the haloperidol-treated patients entered the maintenance phase compared to 61% of the olanzapine-treated patients. A greater percentage of haloperidol-treated patients compared to olanzapine-treated patients discontinued during the maintenance phase and data were not collected after discontinuation. Therefore, olanzapine-treated patients had a greater opportunity to experience an inpatient stay compared to haloperidol-treated patients. Despite these apparent biases against the olanzapine group, the haloperidol group did not demonstrate greater improvements in work and social outcomes or less time to an inpatient stay compared to the olanzapine group.

One should also note that the results of this study come in the framework of a large randomized clinical trial where the primary endpoint was efficacy, not functional improvement. While such a framework is advantageous in terms of minimizing bias and ensuring that treatment groups are comparable, replication of these findings should be sought in naturalistic studies where compliance rates may be more likely to vary and where the impact of treatment on functioning can be examined at differing levels of disease severity.

The findings of this study indicate that olanzapine's superior clinical profile leads to differential improvements in social and vocational functioning beyond what is achieved with typical antipsychotic treatment. In addition, acute phase findings showing differential improvements for olanzapine-treated patients in clinical response, social functioning, and day-to-day life participation compared to haloperidol-treated patients provides further evidence for the relationship between negative symptom efficacy and social function. While enhanced symptom control appears to lead to increased levels of functioning, it may also allow patients to engage more fully in rehabilitation initiatives. Indeed, the impact of olanzapine on reducing hospitalization and improving functioning may not have been fully realized in this study, since the use of atypical antipsychotics in combination with a psychiatric rehabilitation programme may lead to greater improvements in functioning (Littrell, 1995, 1996; Littrell and Littrell, 1997, 1998). Nonetheless, the potential for functional improvement seen with olanzapine and other atypical antipsychotics provide important evidence for first-line use of these agents and will expand treatment objectives from controlling psychosis to reintegrating the patient into the community.

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